Exercise 1

On differentiation of trigonometric functions



Interactive test

From the school book | Understand | Apply | Higher Order Thinking Skills

1 **Find the first derivative for each of the following :**

(1) $y = \cot 5x$

(2) $y = \sec 7x$

(3) $y = \csc \frac{x}{2}$

(4) $y = \sec (5x + 2)$

(5) $y = \csc (\pi - 2x)$

(6) $y = \cot (x^2 + 3)$

(7) $y = x + \cot 2x$

(8) $y = \sqrt[3]{x} + \sec 5x$

(9) $y = \cot \left(\frac{1}{2} x^4\right)$

(10) $y = \sec (3x^4 + 2x^2 - 1)$

(11) $y = \cot \left(\pi - \frac{1}{x}\right)$

(12) $y = 3x^5 + 4 \cot x$

(13) $y = \cos 2x - 5 \cot 3x$

(14) $y = 3 \sec 2x + 2 \csc 3x$

(15) $y = \sec \frac{x}{4} + \sec \frac{\pi}{4}$

(16) $y = 3 \csc \sqrt{1 + x^2}$

2 **Find the first derivative for each of the following :**

(1) $y = x \sec 3x$

(2) $y = x^2 \cot 3x$

(3) $y = \sec x \tan 2x$

(4) $y = \csc 2x \sec 3x$

(5) $y = (x^2 + 5x + 6) \sec x$

(6) $y = \sec x^2 \tan x^2$

(7) $y = x^2 \sec \frac{1}{x}$

(8) $y = (3x + 4)(5x + \csc x)$

(9) $y = \sqrt{x} \sec \sqrt{x}$

(10) $y = x^2 \cos x \cot x$

3 **Find $\frac{dy}{dx}$ for each of the following :**

(1) $y = \frac{\cot x}{x}$

(2) $y = \frac{\csc x}{x}$

(3) $y = \frac{6 - x}{\sec x}$

(4) $y = \frac{\cot 3x}{2x + 3}$

(5) $y = \frac{\csc 2x}{x^2}$

(6) $y = \frac{1 - \sec x}{1 + \sec x}$

(7) $y = \frac{\sec x}{1 + \sec x}$

(8) $y = \frac{\csc (3x + 1)}{\tan (4x - 1)}$

4 **Find the first derivative for each of the following :**

(1) $y = \csc^4 x$

(2) $y = \cot^7 x$

(3) 📖 $y = 3 \sec^2 (2x + \pi)$

(4) 📖 $y = \csc^2 (1 + 3x^2)$

(5) 📖 $y = \cot^3 \sqrt{x}$

(6) 📖 $y = \sin^2 3x + \sec^2 x$

(7) $y = x \cot^2 x$

(8) $y = x^3 \sec^2 3x$

(9) $y = \dfrac{\sec^2 x}{x}$

(10) $y = \dfrac{x}{\csc^2 2x}$

(11) $y = \cot 7x^2 + \cot^2 7x$

(12) $y = \sec^3 2x \tan^3 2x$

(13) $y = \sqrt{1 + \cot x}$

(14) 📖 $y = \sqrt{1 + \csc x}$

(15) $y = \sqrt[3]{x + \cot x}$

(16) $y = \sqrt{x^3 + \cot^2 4x}$

5 **Find the first derivative for each of the following :**

(1) 📖 $y = (1 + \cot x)^2$

(2) 📖 $y = (3 - 2 \cot x)^2$

(3) 📖 $y = (\csc x + \cot x)^{-1}$

(4) $y = \dfrac{1}{(x + \csc x)^3}$

(5) 📖 $y = \tan (\cot x)$

(6) 📖 $y = \cot (\cos 3x)$

(7) $y = \sec (\tan^2 x)$

(8) $y = \sec^2 (\tan x)$

(9) 📖 $y = \sin (\csc 3x^2)$

(10) $y = \tan (\sec x^2)$

6 **Find the slope of the tangent to the curve of the function f where $y = f(x)$ for each of the following at the indicated points :**

(1) $y = \csc \dfrac{x}{2}$ at $x = \dfrac{\pi}{2}$

(2) 📖 $y = 2 \cot x + \sqrt{2} \sec x$ at $x = \dfrac{\pi}{4}$

(3) 📖 $y = 3 \tan x - \csc^2 x$ at $x = \dfrac{3\pi}{4}$

(4) $y = 3 \cot^5 x$ at $x = \dfrac{\pi}{4}$

7 Choose the correct answer from the given ones :

(1) If $f(x) = \cot(5x - \pi)$, then $\grave{f}\left(\frac{\pi}{4}\right) =$ .....

(a) $5\sqrt{2}$ (b) $-5\sqrt{2}$ (c) 10 (d) – 10

(2) If $y = \csc 2x$, then $\frac{dy}{dx} =$ ..... at $x = \frac{\pi}{6}$

(a) $-\frac{3}{4}$ (b) $\frac{-4}{3}$ (c) $-\frac{1}{2}$ (d) $\sqrt{3}$

(3) If $f(x) = \sec^2 x - \tan^2 x$, then $\grave{f}(-1) =$ .....

(a) – 1 (b) zero (c) 1 (d) 2

(4) If $f(x) = (5 - 2\cot x)^3$, then $\grave{f}\left(\frac{\pi}{4}\right) =$ .....

(a) – 108 (b) – 54 (c) 54 (d) 108

(5) If $y = \tan x + \frac{1}{3}\tan^3 x$, then $\frac{dy}{dx} = (\text{.....})^4$

(a) $\csc x$ (b) $\sec x$ (c) $\tan x$ (d) $\sec^4 x$

(6) If $f(x) = \sin x \sec 2x$, then $\grave{f}(0) =$ .....

(a) – 1 (b) zero (c) 1 (d) 2

(7) $\frac{d}{dx}\left(2\cot\frac{\pi}{4}\right) =$ .....

(a) $-2\csc^2\frac{\pi}{4}$ (b) $-2\cot\frac{\pi}{4}\csc\frac{\pi}{4}$ (c) 2 (d) zero

(8) $\frac{d}{dx}(2\cos^2 x - 1) =$ .....

(a) $\sin 2x$ (b) $\cos 2x$ (c) $-2\sin 2x$ (d) $-2\cos 2x$

(9) $\frac{d}{dx}(1 + \tan^2 x)^3 =$ .....

(a) $6\sec^5 x \tan x$ (b) $3\sec^2 x \tan x$ (c) $6\sec^6 x \tan x$ (d) $3\sec^4 x \tan x$

(10) $\frac{d}{dx}\left[x^2 + \frac{d}{dx}(x + \sec x)\right] =$ .....

(a) $2x + 1 + \sec x \tan x$ (b) $2x + 2\sec^3 x - \sec x$

(c) $2 + \sec^3 x - \sec x$ (d) $2x + \sec^2 x \tan^2 x$

(11) If $f(x) = 8x \sin x \cos x \cos 2x$, then $\grave{f}\left(\frac{\pi}{8}\right) =$ .....

(a) – 2 (b) zero (c) 1 (d) 2

(12) $\frac{d}{dx}\left[(\sec x - 1)(\sec x + 1)\right] =$ .....

(a) $\sec^2 x \tan^2 x$ (b) $2\sec^2 x \tan x$ (c) $\sec^2 x \tan x$ (d) $\sec^4 x$

(13) $\frac{d}{dx}(\cos x \csc x) =$ .....

(a) zero (b) $\csc^2 x$ (c) $\sec^2 x$ (d) $-\csc^2 x$

(14) If $y = \sin x \sec\left(\frac{\pi}{2} - x\right)$ where x is an acute angle , then $\frac{dy}{dx} =$ .....

(a) zero (b) 1

(c) $\cos x \csc x + \sin x \sec x$ (d) $\sin x \cos x + \sec x \cot x$

(15) If $y = \cos\left(\frac{\pi}{2} - \frac{1}{x}\right)$, then $\frac{dy}{dx} =$ .....

(a) $-\cos\frac{1}{x}$ (b) $\frac{1}{x^2}\cos\frac{1}{x}$

(c) $\frac{-1}{x^2}\sin\left(\frac{\pi}{2} - \frac{1}{x}\right)$ (d) $-\frac{1}{x^2}\cos\left(\frac{\pi}{2} - \frac{1}{x}\right)$

(16) If $y = \sec\left(\frac{\pi}{4}x\right)$, then $\frac{dy}{dx} =$ .....

(a) $\frac{\pi}{4}\sec\left(\frac{\pi}{4}x\right)\tan\left(\frac{\pi}{4}x\right)$ (b) $-\frac{\pi}{4}\sec\left(\frac{\pi}{4}x\right)\tan\left(\frac{\pi}{4}x\right)$

(c) $\frac{\pi}{4}\sec\left(\frac{\pi}{4}\right)\tan\left(\frac{\pi}{4}\right)$ (d) $\sec\left(\frac{\pi}{4}x\right)\tan\left(\frac{\pi}{4}x\right)$

(17) If $y = \left(\sec\frac{\pi}{4}\right)x$, then $\frac{dy}{dx} =$ .....

(a) $\frac{\pi}{4}\sec\left(\frac{\pi}{4}x\right)\tan\left(\frac{\pi}{4}x\right)$ (b) $\frac{\pi}{4}x$

(c) $\frac{\pi}{4}$ (d) $\sqrt{2}$

(18) If $f(x) = \sec x \cos x + \csc x \sin x$, then $f\left(\frac{13\pi}{4}\right) + \grave{f}\left(\frac{13\pi}{4}\right) =$ .....

(a) $\frac{-13\pi}{4}$ (b) $\frac{13\pi}{4}$ (c) 2 (d) undefined.

(19) If $y + \cot x = 0$, then $\frac{dy}{dx} =$ .....

(a) $1 + y$ (b) $1 + y^2$ (c) $y^2 - 1$ (d) y^2

(20) *(2nd session 2022)* If $y = (1 + \tan x)\cot x$, then $\frac{dy}{dx} =$ .....

(a) $\csc^2 x$ (b) $\sec^2 x$ (c) $-\sec^2 x$ (d) $-\csc^2 x$

(21) *(1st session 2022)* If $y = \sec^2 2x$ where $x \in]0, \frac{\pi}{4}[$, then $\frac{2y}{\grave{y}} =$ .....

(a) $\frac{1}{2}\cot 2x$ (b) $\cot 2x$ (c) $2\tan 2x$ (d) $\tan 2x$

(22) If $y = \frac{1}{4}\sec^4 x$, then $\frac{dy}{dx} =$ .....

(a) $4y \tan x$ (b) $4y$ (c) $\sec^3 x \tan x$ (d) $y \tan x$

(23) If $y = (\csc X - \cot X)(\csc X + \cot X)$, then $\frac{dy}{dX} =$ .....

(a) zero (b) 1 (c) $-y \csc X$ (d) $-y \cot X$

(24) If $y = \frac{2 \cot X}{\cot^2 X - 1}$, then $\frac{dy}{dX} =$ ..... where $X \in]0 , \frac{\pi}{6}[$

(a) $2 \sec^2 2X$ (b) $2 \cot^2 2X$ (c) $4 \csc^2 2X$ (d) $\tan 2X$

(25) If $y = \sec 3X + \tan 3X$, then $\frac{dy}{dX} =$ .....

(a) $3y \sec 3X$ (b) $3y \tan 3X$ (c) $3y \sec^2 3X$ (d) $3y \tan^2 3X$

(26) If $y = \cot aX$ and $\frac{dy}{dX} + 4(1 + y^2) = 0$, then $a =$ .....

(a) 1 (b) -2 (c) 4 (d) -4

(27) If $y = \sin^2 3X + \cos^2 3X + \cot^2 3X$ and $\frac{dy}{dX} = a \csc^2 3X \cot 3X$, then $a =$ .....

(a) -1 (b) -2 (c) -3 (d) -6

(28) If $y = \frac{\sec X \csc X}{\csc^2 X - \sec^2 X}$, then $\frac{dy}{dX} =$ .....

(a) $\sec^2 (2X)$ (b) $\frac{1}{2} \tan (2X)$ (c) $\sec X \csc X$ (d) $2 \sin^2 (2X)$

(29) If $f(X) = \tan (X - \theta) \cdot \cot (X + \theta)$, then $f'(X) =$ ..... at $X = \theta$

(a) $\cot \theta$ (b) $\tan 2\theta$ (c) $\cot 2\theta$ (d) $\tan \theta + \cot \theta$

(30) The slope of the curve : $y = \sqrt{X + \sec X}$ at $X = 0$ equals .....

(a) -1 (b) zero (c) $\frac{1}{2}$ (d) 1

(31) $\lim_{h \to 0} \frac{\cot (X + h) - \cot (X)}{h} =$ .....

(a) $-\csc^2 X$ (b) $\sec^2 X$ (c) $\cot^2 X$ (d) $\tan^2 X$

(32) $\lim_{h \to 0} \frac{\sec \left(\frac{\pi}{4} + h\right) - \sec \left(\frac{\pi}{4}\right)}{h} =$ .....

(a) $\sqrt{2}$ (b) zero (c) $\frac{1}{\sqrt{2}}$ (d) not exist.

(33) If $y = \frac{1 + \tan X}{1 - \tan X}$, then $\frac{dy}{dX} =$ .....

(a) $\cos^2 (X + 45°)$ (b) $\sec^2 (X + 45°)$ (c) $\sin^2 (X + 45°)$ (d) $\csc^2 (X + 45°)$

(34) *(1st session 2021)* If $y = \sqrt{2 (\sec X + \tan X)}$ where $X \in]0 , \frac{\pi}{2}[$, then $\frac{1}{y} \left(\frac{dy}{dX}\right) =$ .....

(a) $2 \sec X$ (b) $-\frac{1}{2} \sec X$ (c) $\frac{1}{2} \sec X$ (d) $-2 \sec X$

8 If $y = \cot \frac{\pi}{6} z$, $z = 3\sqrt{x}$

Find : $\frac{dy}{dx}$ at $x = 1$

9 If $y = \sin x + \sec x$, $x = 3\pi z$

Find : $\frac{dy}{dz}$ at $z = 1$

10 If $y = \sqrt{5 - 2z}$, $z = \sec 2x$

Prove that : $\sqrt{3} \frac{dy}{dx} + 12 = 0$ at $x = \frac{\pi}{6}$

11 If $y = (5z + 1)^5$, $z = \frac{1}{5} \cot 2x$

Find : $\frac{dy}{dx}$ at $x = \frac{\pi}{8}$

12 If $y = \csc x + \cot x$

Prove that : $\frac{dy}{dx} + y \csc x = 0$

13 If $y = (\sec x + \tan x)^n$

Prove that : $\frac{dy}{dx} = n y \sec x$

14 If $y = \sec^n x$

Prove that : $\frac{dy}{dx} = n y \tan x$

15 If $y = \sec x (\sin x + \cos x)$

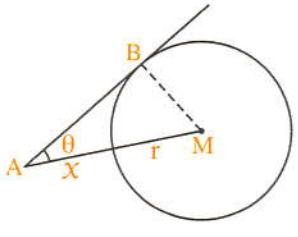
Prove that : $\frac{dy}{dx} - \tan^2 x = 1$

16 If $y = z^4 - 2z^2 + 1$, $z = \csc 4x$

Prove that : $\frac{dy}{dx} + 32 y \csc 8x = 0$

17 **In the opposite figure :**

A is a point moving in the plane , $\overline{AB}$ is a tangent to circle M at B , $AM = x + r$ where r is the radius length of the circle :



(1) Prove that $x = r (\csc \theta - 1)$

(2) Find the rate of change of x with respect to θ when $\theta = \frac{\pi}{6}$

Higher skills

18 **Choose the correct answer from those given :**

(1) If $y = \cot X$, then $\frac{dy}{dX}$ is defined for all values of $X \in$ .....

(a) $\mathbb{R}$ (b) $\mathbb{R} - \{0\}$

(c) $\mathbb{R} - \{n\pi$ where $n \in \mathbb{Z}\}$ (d) $\mathbb{R} - \{\frac{n\pi}{2}$ where $n \in \mathbb{Z}\}$

(2) If $0 < X < \frac{\pi}{2}$ and $f(\tan X) = \sin X$, then $\grave{f}(1) =$ .....

(a) 1 (b) $\frac{\sqrt{2}}{2}$ (c) $\frac{\sqrt{2}}{4}$ (d) $\frac{\sqrt{2}}{6}$

(3) If $\grave{f}(X) = \cos^3 X$ and $f(0) = 0$, then $\lim_{X \to 0} \frac{f(X)}{X} =$ .....

(a) – 1 (b) zero (c) 1 (d) does not exist.

(4) If $y = \sec |X|$, then $\frac{dy}{dX} =$ .....

(a) $\sec |X| \tan |X|$ (b) $-\sec X \tan X$

(c) $\frac{X}{|X|} \sec |X| \tan |X|$ (d) $\frac{X}{|X|} \sec X \tan X$

(5) If $0 < X < \frac{\pi}{2}$ and $f(\sin X) = a \cos X$ where a is a constant and $\grave{f}\left(\frac{3}{5}\right) = -6$, then a = ...

(a) 2 (b) 4 (c) 6 (d) 8

(6) If $y = \cot X$ where X is in degree , then $\frac{dy}{dX} =$ .....

(a) $-\csc^2 X$ (b) $-\csc X \cot X$

(c) $\frac{-\pi}{180} \csc^2 X$ (d) $\frac{-180}{\pi} \csc^2 X$